KPNA7-related Preimplantation Embryonic Arrest — Comprehensive Research Report

Disease: KPNA7-related Preimplantation Embryonic Arrest (PREMBA) **MONDO ID:** MONDO:0957220 · **Disease OMIM:** 620319 (oocyte/zygote/embryo maturation arrest 17; **OZEMA17**) **Causal gene: KPNA7** (karyopherin subunit alpha 7 / importin-α8) — HGNC:21839 · gene OMIM:614107 · locus 7q22.1 **Category:** Mendelian (autosomal recessive, maternal-effect)

Summary

KPNA7-related preimplantation embryonic arrest (PREMBA) is an ultra-rare, autosomal-recessive, maternal-effect female-infertility disorder caused by biallelic loss-of-function variants in *KPNA7*, the gene encoding the oocyte- and early-embryo-restricted nuclear-import receptor karyopherin subunit α7 (importin-α8). The causal gene is **KPNA7** — stated explicitly here because the disease is catalogued under the serial-number synonym OZEMA17 (oocyte maturation arrest 17), which does not name the gene. The disease was defined in 2023 by whole-exome sequencing (WES) burden testing in a cohort of 606 Chinese women with PREMBA versus 2,813 controls, which pinpointed *KPNA7* and 10 affected families carrying rare biallelic variants, most commonly the recurrent missense **c.607C>T (p.Leu203Phe/L203F)** (PMID: 36647821).

The core pathophysiology is a failure of **nucleo-cytoplasmic transport during the oocyte-to-embryo transition**. KPNA7 is a maternally supplied importin-α whose expression is almost entirely restricted to oocytes and cleavage-stage embryos. Pathogenic variants reduce KPNA7 protein levels and impair its ability to bind and nuclear-import its cargo **RSL1D1** (ribosomal L1 domain-containing protein 1). Because the correct nuclear delivery of maternal transcriptional/chromatin-remodeling factors is required to switch on the embryo's own genome, the import defect blocks **zygotic genome activation (ZGA)**. Morphologically normal oocytes fertilize normally and begin cleavage, but embryos arrest before the 8-cell stage (day 3) or blastocyst stage (day 5), leaving no transferable embryos and producing recurrent IVF/ICSI failure.

The mechanism is strongly corroborated in a mouse model. Because mouse *Kpna7* is **not** the functional equivalent, the **mouse functional ortholog is *Kpna2***. *Kpna2*$^{-/-}$ female mice recapitulate PREMBA with 2-cell-stage arrest and ZGA failure, and — critically — the arrest is rescued by microinjection of **human *KPNA7* cRNA but not mouse *Kpna7* cRNA**, establishing cross-species functional equivalence. Oocyte-specific *Rsl1d1* knockout reproduces the same phenotype, nailing down the cargo. There is currently **no cure or pharmacotherapy**; affected women are otherwise systemically healthy, and **donor-oocyte IVF** is the established route to a live birth.

Key Findings

Finding 1 — *KPNA7* is the causal gene for PREMBA, identified by exome burden testing

A population- and gene-based burden test on whole-exome sequencing from **606 women with PREMBA vs 2,813 controls** identified *karyopherin subunit α7 (KPNA7)* as the candidate gene, with rare biallelic/heterozygous variants significantly enriched in cases. The disease is catalogued as **OMIM:620319** (oocyte/zygote/embryo maturation arrest 17, OZEMA17).

> *"using whole-exome sequencing data from 606 women experiencing PREMBA compared with 2,813 controls, we performed a population and gene-based burden test and identified a candidate gene, karyopherin subunit α7 (KPNA7)"* — PMID: 36647821

This is the foundational human-genetics evidence (evidence type: **human clinical / case-control exome study**). The signal was statistically robust (burden $P = 2.88\times10^{-8}$ for rare biallelic variants, below the Bonferroni threshold $P = 1.25\times10^{-5}$).

Finding 2 — Mechanism: variants impair RSL1D1 binding and nuclear import, disrupting ZGA

In vitro, PREMBA-associated KPNA7 variants **reduced KPNA7 protein levels**, **impaired binding to the cargo RSL1D1**, and **reduced nuclear-transport activity**. In vivo, *Kpna2*$^{-/-}$ female mice (mouse ortholog of human KPNA7) show embryo arrest driven by zygotic genome activation defects, and oocyte-specific *Rsl1d1* knockout recapitulates this.

> *"In vitro studies showed that identified sequence variants reduced KPNA7 protein levels, impaired KPNA7 capacity for binding to its substrate ribosomal L1 domain-containing protein 1 (RSL1D1), and affected KPNA7 nuclear transport activity"* — PMID: 36647821
>
> *"Kpna2-/- female mice showed embryo arrest due to zygotic genome activation defects, recapitulating the phenotype of human PREMBA"* — PMID: 36647821

The chain is therefore: loss-of-function variant → reduced/impaired KPNA7 → failed nuclear import of RSL1D1 (and likely other cargoes) → ZGA failure → embryo arrest (evidence types: **in vitro** + **model organism**).

Finding 3 — KPNA7 is an oocyte/embryo-restricted karyopherin-α with distinct cargo specificity

KPNA7 expression is largely limited to early development (oocytes and cleavage-stage embryos). In pig, *KPNA7* knockdown by interfering RNA significantly lowered embryo cell numbers ($P<0.05$). Biochemically, KPNA7 has atypical behavior: it can bind Importin-β **in the absence of NLS cargo** and maintains an "open" conformation, unlike other KPNA family members whose IBB domain auto-inhibits the NLS-binding groove.

> *"Embryos injected with KPNA7-interfering RNAs possessed significantly lower cell numbers than their respective control groups ($P<0.05$)"* — PMID: 22281085
>
> *"KPNA7, an isoform with expression mostly limited to early development, can bind Importin-β (Imp-β) in the absence of NLS cargo"* — PMID: 31642884

This establishes both the **developmental-window specificity** (why the disease is confined to preimplantation development) and the **unique cargo/NLS specificity** that explains why KPNA7 cannot be fully substituted by other importin-α paralogs.

Finding 4 — Gene identifiers and population constraint: LoF-tolerant, consistent with recessive maternal-effect inheritance

Attribute	Value
Gene symbol	KPNA7
Aliases	IPOA8, OZEMA17 (importin-α8)
HGNC	HGNC:21839
Entrez Gene	402569
Ensembl	ENSG00000185467
Gene OMIM	614107
Disease OMIM	620319
Cytoband	7q22.1 (GRCh38 chr7:99,173,572–99,250,075, minus strand)
Pseudogene	present on chr5
gnomAD pLI	5.9×10^{-19} (≈0 — LoF-tolerant)
Observed/expected LoF	54/53.7 (o/e 1.01; 90% CI 0.81–1.26)
Missense Z	0.07

The near-zero pLI and o/e(LoF) ≈ 1 indicate *KPNA7* is **tolerant of heterozygous loss of function** at the population level — exactly what is expected for a **recessive, maternal-effect** gene whose phenotype manifests only in the embryos of homozygous/compound-heterozygous women and has no effect on carrier viability or reproduction. (Evidence type: **computational / population genomics**.)

Finding 5 — KPNA7 somatic functions: mitotic fidelity and nuclear-envelope integrity

Although KPNA7 is virtually silent in adult somatic tissues, it is **re-expressed in some cancers**, where it supports proliferation and genome stability. siRNA knockdown in pancreatic and breast cancer cell lines reduced proliferation and S-phase fraction, produced aberrant centrosome numbers and distorted mitotic spindles, reorganized lamin A/C and B1, and produced lobulated/elongated nuclei.

> *"the silencing of KPNA7 results in a dramatic reduction in pancreatic and breast cancer cell growth, irrespective of the endogenous KPNA7 expression level. This growth inhibition is accompanied by a decrease in the fraction of S-phase cells as well as aberrant number of centrosomes and severe distortion of the mitotic spindles"* — PMID: 29580221
>
> *"KPNA7 depletion leads to reorganization of lamin A/C and B1, the main nuclear lamina proteins, and drastic alterations in nuclear morphology"* — PMID: 29580221

These somatic roles (cell-cycle progression, centrosome number, mitotic-spindle integrity, nuclear-lamina organization) are mechanistically relevant: beyond RSL1D1 import, KPNA7 loss in the cleavage-stage embryo could also compromise mitotic fidelity — a plausible contributor to arrest (evidence type: **in vitro, cancer cell lines**).

Finding 6 — Specific pathogenic variants and genotypes (10 families; recurrent p.L203F)

In the discovery cohort, rare biallelic *KPNA7* variants (transcript **NM_001145715.3**) occurred in **10 patients and 0 controls** (burden $P = 2.88\times10^{-8}$). Across the 10 independent families:

Family(ies)	Genotype	Variant(s)
1, 2, 4, 5, 7, 10	Homozygous	c.607C>T, **p.Leu203Phe (L203F)**
3	Compound het	p.L203F + c.635C>T (**p.Pro212Leu**)
6	Compound het	p.L203F + c.523C>A (**p.Gln175Lys**)
9	Compound het	p.L203F + c.1350_1356delGTGTCTT (**p.Cys451***, truncating)

No homozygotes for the recurrent variant existed in gnomAD/ExAC (MAF<1%) or in-house controls. Homozygosity mapping suggested a **low probability of a founder effect** (recurrence of p.L203F likely reflects a mutational hotspot or independent origins rather than a single shared haplotype). All five tested mutant proteins showed reduced KPNA7 levels and impaired RSL1D1 binding/nuclear transport.

> *"In vitro studies showed that identified sequence variants reduced KPNA7 protein levels, impaired KPNA7 capacity for binding to its substrate ribosomal L1 domain-containing protein 1 (RSL1D1), and affected KPNA7 nuclear transport activity"* — PMID: 36647821

Finding 7 — Clinical phenotype, epidemiologic context, and differential diagnosis

The PREMBA phenotype: morphologically normal oocytes are retrieved, fertilize normally, and undergo the first cleavage division, **but most embryos arrest before the 8-cell stage (day 3) or before the blastocyst stage (day 5)**, yielding no viable embryos and recurrent IVF/ICSI failure. About **10% of all IVF/ICSI embryos** arrest at primary cleavage stages (the broad clinical backdrop). Most previously identified PREMBA genes encode **subcortical maternal complex (SCMC)** proteins — *PADI6, NLRP2, NLRP5, KHDC3L* — whereas *KPNA7* defines a **distinct nuclear-transport mechanism**. Biallelic variants in *KPNA7* plus *PADI6* together explained ~**3.6% of the cohort (22/606)**.

> *"Preimplantation embryo arrest (PREMBA) is a common cause of female infertility and recurrent failure of assisted reproductive technology"* — PMID: 36647821

Finding 8 — Mouse *Kpna2* (not *Kpna7*) is the functional ortholog; *Kpna2*$^{-/-}$ recapitulates 2-cell arrest with ZGA failure

Kpna2$^{-/-}$ female mice recapitulate PREMBA: reduced body size, small ovaries, few oocytes on superovulation, and embryos uniformly arrested at the **2-cell stage** (coinciding with major ZGA, which in mouse initiates at the 2-cell stage). Oocyte-specific *Rsl1d1* knockout reproduces the phenotype. Cross-species rescue established the orthology: microinjection of **human *KPNA7* cRNA — but not mouse *Kpna7* — rescued** the *Kpna2*$^{-/-}$ arrest.

> *"complementary RNA (cRNA) microinjection of human KPNA7, but not mouse Kpna7, was able to rescue the embryo arrest phenotype in Kpna2-/- mice, suggesting mouse KPNA2 might be a homologue of human KPNA7"* — PMID: 36647821
>
> *"As shown in embryos of Kpna2-null females, the consequences are disrupted zygotic genome activation and arrest of development"* — PMID: 36647833

This is the single most important cross-species validation (evidence type: **model organism, genetic rescue**) and is a crucial caution for model selection: targeting mouse *Kpna7* would **not** model the human disease.

Finding 9 — Protein architecture: IBB domain + 8 ARM repeats; missense variants map to the NLS-binding ARM groove

KPNA7 (UniProt **A9QM74**) is a **516-aa importin-α**: an N-terminal importin-β-binding (**IBB**) domain (~aa 1–57) and **8 armadillo (ARM) repeats** (~aa 103–439) forming the NLS-binding groove. The PREMBA missense variants map into the cargo-binding region: **p.Gln175Lys** (ARM2, ~146–185), **p.Leu203Phe** and **p.Pro212Leu** (ARM3, ~188–228); the truncating **p.Cys451*** removes the C-terminal/acidic region downstream of ARM8.

This structural mapping provides a direct mechanistic rationale: the recurrent and compound-heterozygous missense substitutions sit in the ARM repeats that build the NLS-binding surface, consistent with the observed **loss of RSL1D1 binding** (evidence type: **computational / structural annotation**).

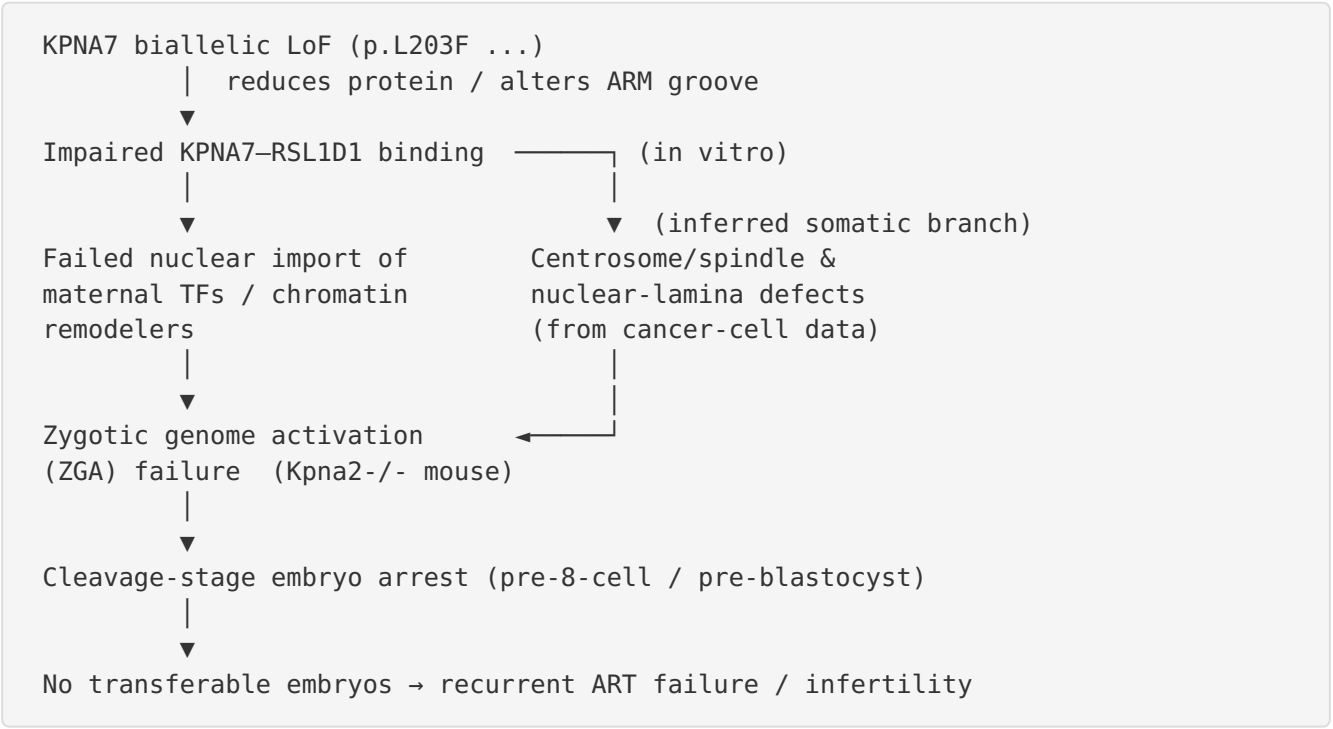
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *KPNA7* loss-of-function variants** (most commonly homozygous c.607C>T, p.Leu203Phe in ARM3) are inherited by the affected woman → **leads to** a mutant, destabilized KPNA7 importin-α in her oocytes.
2. **Reduced KPNA7 protein level + structurally altered NLS-binding ARM groove → results in** impaired binding of KPNA7 to its cargo **RSL1D1** (and probably other NLS-bearing maternal factors). *(Demonstrated in vitro.)*
3. **Loss of KPNA7-mediated nuclear import** of these cargoes → **leads to** failure to deliver transcriptional/chromatin-remodeling factors into the embryonic nucleus at the oocyte-to-embryo transition. *(Inferred from cargo identity + import assays.)*
4. **Nuclear-transport failure → results in** defective **zygotic genome activation (ZGA)** — the embryo cannot switch from maternal to its own transcriptional program. *(Demonstrated in Kpna2-null mouse; inferred in human.)*
5. **ZGA failure → leads to** developmental arrest: in human, before the 8-cell (day 3) / blastocyst (day 5) stage; in mouse, at the 2-cell stage (where major ZGA occurs).
6. **Branch (inferred, from somatic KPNA7 biology):** reduced KPNA7 may additionally compromise **centrosome number, mitotic-spindle integrity, and nuclear-lamina**

organization in cleavage divisions → contributing to arrest independently of or alongside ZGA failure.

7. **Embryo arrest → results in** no transferable embryos → **recurrent IVF/ICSI failure and female infertility** (the clinical phenotype). The woman herself is systemically healthy.



Human-vs-mouse orthology (critical for interpretation and modeling)

Species	Functional gene at oocyte-to-embryo transition	Knockout phenotype	Notes
Human	**KPNA7**	PREMBA (pre-8-cell/ blastocyst arrest)	disease gene
Mouse	**Kpna2** (NOT Kpna7)	2-cell arrest, ZGA failure	rescued by human *KPNA7* cRNA, not mouse *Kpna7*

Ontology suggestions. Biological process (GO): **GO:0006606** (protein import into nucleus), **GO:0051170** (import into nucleus), **GO:0017038** (protein import), positive regulation of embryonic transcription / ZGA. Molecular function (GO): **GO:0008139** (nuclear localization sequence binding), **GO:0061608** (nuclear import signal receptor activity). Cellular component (GO): **GO:0005635** (nuclear envelope), **GO:0005643** (nuclear pore), **GO:0005730** (nucleolus, for RSL1D1), **GO:0005813** (centrosome). Cell types (CL): **CL:0000023** (oocyte), **CL:0000670**

(primordial germ cell), early blastomere/embryonic cell. Anatomy (UBERON): **UBERON:0000992** (ovary), **UBERON:0000922** (embryo), oocyte/zygote structures. Chemical entity (CHEBI): GTP (**CHEBI:15996**, Ran-GTP cycle context). Disease: **MONDO:0957220**.

Section-by-Section Disease Characteristics

1. Disease Information

KPNA7-related PREMBA is a Mendelian female-infertility disorder in which fertilized oocytes arrest during preimplantation cleavage. **Causal gene: KPNA7.** Identifiers: **MONDO:0957220**, **disease OMIM:620319** (OZEMA17), **gene OMIM:614107**. There is no dedicated ICD-10/ICD-11, MeSH, or distinct Orphanet code specific to the KPNA7 subtype in the available evidence; clinically it falls under female infertility / recurrent ART failure. Synonyms: **OZEMA17** (oocyte/zygote/embryo maturation arrest 17); gene aliases **IPOA8** / **importin-α8**. Information is a mix of **individual-patient** data (10 families, in-depth genotype–phenotype) and **aggregate** resources (OMIM, gnomAD).

2. Etiology

Primary cause: genetic — biallelic (homozygous or compound-heterozygous) loss-of-function variants in *KPNA7*. **Genetic risk factor:** inheriting two pathogenic *KPNA7* alleles; the recurrent **p.L203F** is the principal allele. **Consanguinity** plausibly increases homozygosity risk (6/10 families homozygous), though a founder effect was judged unlikely. No environmental, infectious, lifestyle, protective, or gene-environment factors are established — this is a monogenic maternal-effect disorder. Carriers (heterozygotes) are unaffected and fertile, consistent with the LoF-tolerant gnomAD constraint.

3. Phenotypes

The phenotype is reproductive and effectively **embryo-level**, not a systemic phenotype in the affected woman: - **Preimplantation embryonic arrest** (cleavage-stage, before 8-cell/blastocyst). HPO candidates: **HP:0000789** (Infertility), **HP:0008215** (Female infertility), and embryonic-arrest/ART-failure terms. - **Recurrent IVF/ICSI failure** — clinical sign; onset at **reproductive age** when ART is attempted; severity: **severe** (no viable embryos); progression: **stable** (consistent across cycles). - **Normal oocyte retrieval, normal fertilization, normal**

first cleavage — the defect is downstream of fertilization. - **Frequency among affected:** essentially 100% embryo arrest within affected women. - **Quality-of-life impact:** substantial reproductive and psychosocial burden of infertility; no somatic morbidity.

4. Genetic/Molecular Information

Causal gene **KPNA7** (HGNC:21839; 7q22.1). Pathogenic variants (NM_001145715.3): **c.607C>T p.Leu203Phe** (recurrent missense), **c.635C>T p.Pro212Leu**, **c.523C>A p.Gln175Lys**, **c.1350_1356del p.Cys451** *(truncating). Variant classes: missense + one truncating allele. Functional consequence: loss of function (reduced protein, impaired cargo binding and nuclear import). Origin: germline.** Population AF: absent/ultra-rare in gnomAD/ExAC (MAF<1%; no control homozygotes). No modifier genes, epigenetic changes, or chromosomal abnormalities are established for this disorder.

5. Environmental Information

Not applicable. No environmental toxins, lifestyle factors, or infectious agents are implicated. This is a purely monogenic maternal-effect disorder.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** above for the full ordered causal chain. Core pathway: **nucleo-cytoplasmic transport** (importin-α/β–Ran system) → **RSL1D1 nuclear import → zygotic genome activation**. Cellular processes: cell-cycle/cleavage division, possible centrosome/spindle and nuclear-lamina involvement. Protein dysfunction: **loss of function** of an importin-α with destabilization and loss of NLS-binding. Upstream = variant/import failure; downstream = ZGA failure and arrest.

7. Anatomical Structures Affected

- **Organ/tissue:** ovary (oocytes) and the preimplantation embryo; female reproductive system (UBERON:0000992 ovary; UBERON:0000922 embryo).
- **Cell types (CL):** oocyte (CL:0000023), zygote, early blastomeres.
- **Subcellular (GO CC):** nuclear pore/nuclear envelope (GO:0005643/GO:0005635), nucleus, nucleolus (RSL1D1), centrosome (GO:0005813), nuclear lamina.
- **Lateralization:** not applicable (bilateral ovaries; defect is cell-intrinsic).

8. Temporal Development

- **Onset:** the embryonic defect manifests at the **oocyte-to-embryo transition / cleavage stage** (first days post-fertilization). Clinically recognized at **reproductive age** upon ART failure.
- **Progression: stable/consistent** — arrest recurs every cycle; not progressive within the woman.
- **Critical period:** the narrow window of **ZGA** during preimplantation development; no intervention window has been demonstrated.

9. Inheritance and Population

- **Inheritance: autosomal recessive, maternal-effect** (phenotype depends on the mother's genotype, expressed in her embryos).
- **Penetrance:** apparently high/complete for embryo arrest in biallelic women (within the reported families).
- **Epidemiology: ultra-rare**; precise prevalence/incidence unknown. Within an infertility cohort enriched for PREMBA, *KPNA7*+*PADI6* biallelic variants explained ~3.6% (22/606); *KPNA7* alone accounted for 10 families.
- **Founder effect:** judged **low probability** despite recurrent p.L203F.
- **Population:** discovery cohort was **Chinese**; broader ethnic distribution undefined.
- **Sex:** phenotype expressed through **females** (maternal-effect); carrier males are unaffected.

10. Diagnostics

- **Genetic testing is the diagnostic cornerstone: WES** (how the gene was found) or targeted **single-gene / infertility gene-panel** testing for biallelic *KPNA7* variants; WGS also applicable. Differential includes SCMC-gene disorders (*PADI6, NLRP2, NLRP5, KHDC3L, TLE6, OOEP*) and other oocyte/zygote maturation-arrest genes.
- **Clinical/functional clue:** time-lapse embryology documenting normal fertilization and first cleavage followed by **consistent pre-8-cell/pre-blastocyst arrest** across multiple cycles.
- No specific serum/urine biomarker, imaging, or histopathology test exists. Karyotype/CMA typically normal.
- **Screening:** carrier screening in consanguineous families / affected pedigrees; cascade testing of relatives.

11. Outcome / Prognosis

- **Reproductive prognosis with autologous oocytes: very poor** — recurrent total embryo arrest and ART failure.
- **No systemic mortality/morbidity** in the affected woman — she is otherwise healthy; life expectancy unaffected.
- **Favorable reproductive outcome is achievable** via donor-oocyte IVF (bypasses the maternal-effect defect).
- Prognostic factor: the defect is embryo-intrinsic and cycle-independent, so repeating autologous cycles is predictably unsuccessful.

12. Treatment

- **No pharmacotherapy, gene therapy, or targeted therapy exists**; none is established for a maternal-effect oocyte transport defect.
- **Standard of care: donor-oocyte IVF** (NCIT: oocyte donation / in vitro fertilization procedures) — the established route to a live birth, as it supplies wild-type KPNA7 maternal protein.
- **Genetic counseling** is a key component of management.
- Experimental concepts (not clinically available): cytoplasmic/spindle transfer or cRNA/protein supplementation of oocytes — supported conceptually by the mouse cRNA-rescue experiment but not translated to humans.

13. Prevention

- **Genetic counseling** for couples with recurrent total embryo arrest and identified biallelic *KPNA7* variants.
- **Carrier/cascade screening** in affected families and consanguineous couples.
- **Reproductive options:** donor-oocyte IVF. PGT is **not** useful here because the defect is a maternal-effect in the egg cytoplasm, not an embryo-genotype issue that can be selected against.
- No primary/secondary public-health or immunization prevention applies.

14. Other Species / Natural Disease

- **Mouse (NCBI Taxon 10090):** functional ortholog is ***Kpna2*** (not *Kpna7*); *Kpna2*$^{-/-}$ females show 2-cell arrest with ZGA failure. Mouse *Kpna7* exists but does **not** rescue the phenotype.

- **Pig (NCBI Taxon 9823):** *KPNA7* RNAi reduces cleavage-stage embryo cell number (PMID: 22281085).
- No naturally occurring companion-animal/wildlife disease (OMIA) is documented for this gene; no zoonotic relevance.

15. Model Organisms

- **Mouse *Kpna2* knockout** (*Kpna2*$^{-/-}$ females): the best genetic model — **recapitulates PREMBA** (small ovaries, few oocytes, 2-cell arrest, ZGA failure); validated by human-*KPNA7*-cRNA rescue. Resource: MGI.
- **Oocyte-specific *Rsl1d1* knockout** mouse: models the cargo-side of the mechanism.
- **Porcine embryo RNAi** (in vitro/ex vivo): demonstrates conserved requirement in cleavage-stage development.
- **Cancer cell lines** (pancreatic, breast) with *KPNA7* siRNA: not disease models, but reveal KPNA7's mitotic/nuclear-envelope functions.
- **Limitations:** the species difference (mouse *Kpna2* vs human *KPNA7*) complicates naive mouse modeling; human preimplantation biology (timing/stage of arrest, specific cargo repertoire) is not fully captured by mouse.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
36647821	*Karyopherin α deficiency contributes to human preimplantation embryo arrest*	Human genetics + in vitro + mouse	**Primary/landmark**: gene discovery, variants, RSL1D1/ import mechanism, Kpna2 rescue
36647833	*Take your mother's ferry: preimplantation embryo development requires maternal karyopherins...*	Commentary/ review	Independent confirmation of ZGA disruption and arrest in Kpna2-null model
31642884	*Characterization of the Importin-β binding domain in KPNA7*	Biochemistry/ structural	KPNA7's early-development expression and atypical Imp-β binding; "open" state
22281085	*KPNA7... required for porcine embryo development*	Model organism (pig)	Functional requirement in cleavage-stage embryos ($P<0.05$)
29580221	*Depletion of KPNA7 induces mitotic defects and deformation of nuclei in cancer cells*	In vitro (cancer lines)	Somatic roles: mitotic spindle, centrosomes, nuclear lamina
26399853	*Changing expression/subcellular distribution of karyopherins during murine oogenesis*	Model organism (mouse)	Karyopherins dynamically regulated in oogenesis; supports KPN roles in oocyte development
34362424	*RSL1D1 coordinates with HDM2 to regulate p53...*	In vitro	Background on cargo RSL1D1 (nucleolar protein, cell-proliferation role)
17158916	*Multiple controls regulate nucleostemin partitioning...*	In vitro	RSL1D1 biology (ribosomal L1 domain; nucleolar partitioning)

How the evidence converges: The landmark paper (PMID: 36647821) supplies the human genetics, the in-vitro mechanism, and the mouse model in one study — an unusually complete causal demonstration. Independent commentary (PMID: 36647833) and prior functional work

in pig (PMID: 22281085) and biochemistry (PMID: 31642884) corroborate the developmental restriction and transport function. The cancer-cell work (PMID: 29580221) adds a plausible secondary (mitotic/nuclear-envelope) mechanism.

Limitations and Knowledge Gaps

- **Single discovery cohort, one ethnicity:** The human genetics derives almost entirely from one Chinese cohort (606 cases). Prevalence, pan-ethnic allele spectrum, and whether p.L203F recurs globally are unknown.
- **Small family count:** 10 families define the full known mutational spectrum; the complete variant landscape and any genotype–phenotype gradation (e.g., day-3 vs day-5 arrest) are under-characterized.
- **Cargo repertoire incomplete:** RSL1D1 is the demonstrated cargo, but KPNA7 likely imports additional maternal factors; the full set and which are rate-limiting for ZGA are unresolved.
- **Secondary mechanism inferred:** The mitotic-fidelity/nuclear-lamina branch is extrapolated from cancer cells, not demonstrated in human embryos.
- **Human-vs-mouse orthology complexity:** Conclusions about ZGA timing/arrest stage rest on mouse *Kpna2*; human preimplantation specifics are inferred.
- **No therapeutics:** No treatment addresses the oocyte defect; only donor oocytes bypass it. No biomarker exists for pre-ART risk stratification.
- **Ontology coverage:** Disease lacks dedicated ICD/MeSH codes; HPO annotation for "embryonic arrest" is imperfect.

Proposed Follow-up Experiments / Actions

1. **Replicate and expand the genetics:** Sequence *KPNA7* in independent, multi-ethnic PREMBA/recurrent-ART-failure cohorts to estimate true prevalence, confirm p.L203F recurrence, and expand the variant spectrum; deposit variants in ClinVar with ACMG classifications.

2. **Define the full KPNA7 cargo set** in human/mouse oocytes (e.g., proximity labeling / IP-MS of KPNA7 across the oocyte-to-embryo transition) to identify which imported factors are rate-limiting for ZGA.
3. **Human-relevant functional modeling:** Use human iPSC-derived or patient oocyte/embryo systems with single-embryo RNA-seq to directly test ZGA failure in human KPNA7-deficient embryos and map the transcriptional defect.
4. **Structural validation:** Solve/compute KPNA7–NLS/RSL1D1 complex structures (AlphaFold-Multimer + biophysics) to confirm how p.L203F/p.P212L/p.Q175K disrupt the ARM NLS groove and quantify binding-affinity changes.
5. **Test the secondary mitotic branch:** Assess centrosome number, spindle integrity, and nuclear-lamina organization in KPNA7-deficient cleavage-stage embryos to determine whether it contributes to arrest independently of ZGA.
6. **Therapeutic proof-of-concept:** Explore oocyte **cRNA/protein supplementation** of wild-type KPNA7 (building on the mouse cRNA rescue) as a conceptual rescue, while recognizing clinical/ethical barriers.
7. **Clinical pathway:** Establish genetic-counseling and **donor-oocyte** referral guidance for women with biallelic *KPNA7* variants and recurrent total embryo arrest; formalize cascade carrier screening in affected families.

Evidence-source legend: human clinical/genetics (PMID: 36647821); model organism — mouse (PMID: 36647821, PMID: 36647833, PMID: 26399853), pig (PMID: 22281085); in vitro/biochemistry (PMID: 31642884, PMID: 29580221, PMID: 34362424, PMID: 17158916); computational/population (gnomAD constraint).
